

# Viacheslav Klimenko

Department of Physics and Astronomy  
University of South Carolina  
712 Main Street, Columbia, SC 29208  
Email: [klimenkv@mailbox.sc.edu](mailto:klimenkv@mailbox.sc.edu)  
Phone: 803-777-0841

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Appointments | <b>Postdoctoral Fellow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  | Oct. 2022 – Present   |
|              | Department of Physics and Astronomy, University of South Carolina, Columbia, SC, USA                                                                                                                                                                                                                                                                                                                                                                        |                       |
|              | <b>Postdoctoral Fellow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  | Apr. 2022 – Sep. 2022 |
|              | Department of Physics and Astronomy, University of South Carolina, Columbia, SC, USA                                                                                                                                                                                                                                                                                                                                                                        |                       |
|              | <b>Research Scientist</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2016 – Apr. 2022      |
|              | Department of Applied Mathematics and Mathematical Physics, Ioffe Institute, St. Petersburg, Russia                                                                                                                                                                                                                                                                                                                                                         |                       |
| Education    | <b>Assistant Professor</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2016 – 2021           |
|              | Department of Astronomy, St. Petersburg Academic University, Russian Academy of Sciences, St. Petersburg, Russia                                                                                                                                                                                                                                                                                                                                            |                       |
|              | <b>Junior Research Scientist</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | 2011 – 2016           |
|              | Department of Theoretical Astrophysics, Ioffe Institute, St. Petersburg, Russia                                                                                                                                                                                                                                                                                                                                                                             |                       |
| Education    | <b>Ph.D. in Astronomy &amp; Astrophysics</b>                                                                                                                                                                                                                                                                                                                                                                                                                | 2016                  |
|              | Ioffe Institute, St. Petersburg, Russia                                                                                                                                                                                                                                                                                                                                                                                                                     |                       |
|              | <b>M.Sc. in Theoretical Physics</b>                                                                                                                                                                                                                                                                                                                                                                                                                         | 2011                  |
|              | St. Petersburg Academic University, St. Petersburg, Russia                                                                                                                                                                                                                                                                                                                                                                                                  |                       |
| Education    | <b>B.Sc. in Physics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2009                  |
|              | Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia                                                                                                                                                                                                                                                                                                                                                                               |                       |
| Summary      | <b>Projects Funding:</b> Total of \$3M in external awards as PI/Co-I/Collaborator (~\$800K as PI and Co-I at USC).                                                                                                                                                                                                                                                                                                                                          |                       |
|              | <b>Publications:</b> 25 refereed manuscripts published in or submitted to peer-reviewed journals                                                                                                                                                                                                                                                                                                                                                            |                       |
|              | - 14 first-author peer-reviewed manuscripts ( <i>ApJ</i> , <i>MNRAS</i> ).                                                                                                                                                                                                                                                                                                                                                                                  |                       |
|              | - 2 best scientific work of the year ( <i>Astronomy Letters Journal</i> , <i>Ioffe Institute</i> ).                                                                                                                                                                                                                                                                                                                                                         |                       |
|              | - Invited to be a referee for <i>Monthly Notices of the Royal Astronomical Society</i> , <i>Astronomy &amp; Astrophysics</i> , <i>Astrophysical Journal</i> , <i>Physics Conference Series</i> .                                                                                                                                                                                                                                                            |                       |
|              | <b>Project management:</b> Lead and participate in several international projects                                                                                                                                                                                                                                                                                                                                                                           |                       |
|              | - probing the gas in and around local galaxies mapped with Integral Field Spectroscopy                                                                                                                                                                                                                                                                                                                                                                      |                       |
|              | - evolution of dust composition and dust extinction across cosmic time                                                                                                                                                                                                                                                                                                                                                                                      |                       |
|              | - comprehensive view of molecular gas in distant galaxies                                                                                                                                                                                                                                                                                                                                                                                                   |                       |
|              | - isotopic abundance ratios and fundamental physics at cosmic noon                                                                                                                                                                                                                                                                                                                                                                                          |                       |
| Summary      | <b>Observing Experience:</b> PI and Co-I of 14 approved proposals using 2–10m class telescopes from ultraviolet to mid-infrared.                                                                                                                                                                                                                                                                                                                            |                       |
|              | <b>Mentoring:</b> 3 graduate students, and 4 undergraduate students.                                                                                                                                                                                                                                                                                                                                                                                        |                       |
|              | <b>Teaching:</b> Teaching experience ~ 10 years. 2 undergraduate/graduate-level courses as Assistant Professor and 2 as Instructor.                                                                                                                                                                                                                                                                                                                         |                       |
|              | <b>Public Outreach:</b> Public lectures at the St. Petersburg Planetarium; <i>Astronomy on Tap</i> ; STEM Hero Fair at USC; Midway Physics Day at the South Carolina State Fair; and Discover USC.                                                                                                                                                                                                                                                          |                       |
|              | <b>Diversity and Inclusion:</b> I am a first-generation scholar from a low-income, nontraditional background. I pursued a path in academic research, established an independent research program, and built collaborations with researchers around the globe from diverse cultural, national, and educational backgrounds. I also value a balanced life that integrates research, family, and outdoor activities (e.g., mountaineering, climbing, running). |                       |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |

|                     |                                                                                                                                                                                                                                                                                                                                                       |         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Teaching Experience | <b>Guest Lecturer</b><br>University of South Carolina, Columbia, SC<br>– ASTR201: Introduction to Astronomy: The Dark Universe - <i>Spring</i> 2025<br>– ASTR 301: Galactic Astrophysics- <i>Spring</i> 2026                                                                                                                                          |         |
|                     | <b>Assistant Professor</b><br>Department of Astronomy, St. Petersburg Academic University, St. Petersburg, Russia<br>Statistical Mechanics - <i>Fall, Spring</i> 2016–2021<br>Introduction to Modern Astronomy - <i>Spring</i> 2019                                                                                                                   |         |
|                     | <b>Teaching Specialist</b><br>Peter the Great St. Petersburg Polytechnic University, St. Petersburg, Russia<br>General physics - <i>Fall, Spring</i> 2011–2014                                                                                                                                                                                        |         |
| Mentoring           | <b>Graduate students:</b><br>Ioffe Institute - Kurichin, Oleg (2019-2022, comentoring with A. Ivanchik)<br>University of South Carolina<br>- Putha, Karthik (2023-present, comentoring with V. Kulkarni)<br>- Santhigiri, Ananthavishnu (2024-present, comentoring with V. Kulkarni)                                                                  |         |
|                     | <b>Undergraduate students:</b><br>University of South Carolina<br>- Lee, Jack (Spring 2024-Fall 2025)<br>- David, Hoffman (Fall 2025-present, comentoring with V. Kulkarni)<br>St. Petersburg Academic University,<br>- Shenyavsky, Vsevolod (2020-2022)<br>Peter the Great St. Petersburg Polytechnic University,<br>- Baryshnikov, Ilya (2020-2022) |         |
| Research Funding    | As a Principal Investigator:                                                                                                                                                                                                                                                                                                                          |         |
|                     | <b>NASA</b> – Probing Dust Extinction Curve in the Milky Way’s Thick Disk and Halo (co-PI), 2027–2030                                                                                                                                                                                                                                                 | ~\$600K |
|                     | <b>NASA</b> – Keck Principal Investigator Data Award, 2026                                                                                                                                                                                                                                                                                            | \$15K   |
|                     | <b>Russian Foundation for Basic Research</b> – Interstellar Ultraviolet Radiation in Distant Galaxies, 2018–2019                                                                                                                                                                                                                                      | \$15K   |
|                     | <b>Dynasty Foundation</b> – Physical Conditions and Chemistry of the ISM/IGM at High Redshift, 2014                                                                                                                                                                                                                                                   | \$10K   |
|                     | As a Co-investigator/Collaborator                                                                                                                                                                                                                                                                                                                     |         |
|                     | <b>JWST Cycle 3</b> – Mining for DIAMONDS (Dust, Ice, And MOlecules iN Distant GalaxieS) over the Past ~7 Billion Years (Co-I), 2024–2027                                                                                                                                                                                                             | \$244K  |
|                     | <b>JWST Cycle 3</b> – Probing the Dust in the Diffuse ISM in Galaxies over the Past ~8 Billion Years with Spectroscopy of Background Quasars (Co-I), 2024–2027                                                                                                                                                                                        | \$264K  |
|                     | <b>NASA</b> – Mapping the Baryon Cycle In and Around Galaxies Through Cosmic Time (collaborator), 2025–2028                                                                                                                                                                                                                                           | \$558K  |
|                     | <b>NASA</b> – Characterizing Interstellar Dust Extinction in Galaxies and Its Cosmic Evolution (collaborator), 2024–2027                                                                                                                                                                                                                              | \$397K  |
|                     | <b>Russian Science Foundation</b> – Physics of the Early Universe: Observational Cosmology (Co-I), 2018–2022                                                                                                                                                                                                                                          | \$400K  |
|                     | <b>Russian Foundation for Basic Research</b> – Molecular and Diffuse ISM at High Redshifts (Co-I), 2018–2019                                                                                                                                                                                                                                          | \$29K   |
|                     | <b>Russian Science Foundation</b> – Physics of the Early Universe: Chemistry and Physical Conditions (Co-I), 2014–2016                                                                                                                                                                                                                                | \$297K  |
|                     | <b>Russian Foundation President Programme</b> – Galaxies in the Early Universe (Co-I), 2013–2014                                                                                                                                                                                                                                                      | \$32K   |
|                     | <b>Russian Foundation for Basic Research</b> – Interstellar Molecular Hydrogen Clouds at Early Stages (Co-I), 2012–2013                                                                                                                                                                                                                               | \$21K   |
|                     | <b>Russian Foundation for Basic Research</b> – Chemistry of the ISM at High Redshift (Co-I), 2011–2012                                                                                                                                                                                                                                                | \$61K   |

|                   |                                                                                                                                                                                     |                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Telescope<br>Time | As a Principal Investigator:                                                                                                                                                        |                 |
|                   | <b>VLT/X-shooter</b> – “SUENO: Survey of Unexplored Extinction in Normal galaxies with X-shooter Observations”, PI: V. Klimenko                                                     | 26.6h, 2026     |
|                   | <b>Keck/NIRES</b> - “The Keck PRISM Survey: Probing Reddening in Intervening Silicate Dust with Near-Infrared Spectroscopy”, PI: V. Klimenko                                        | 1 night, 2026   |
|                   | <b>SAO/SCORPIO</b> - “Observing the emission of ionized gas in galaxies with a strong halo”, PI: V. Klimenko                                                                        | 25h, 2025       |
|                   | As a Co-investigator                                                                                                                                                                |                 |
|                   | <b>Gemini/GNIRS</b> – “Unveiling the Nature of DLA Host Galaxies: NIR Spectroscopy of Nebular Emission at $z \sim 2$ ”, PI: A. Shaban, (Co-I)                                       | 20.4h, 2026     |
|                   | <b>HST/STIS</b> – “Multiwavelength Extinction Curves in Distant Galaxies”, PI: M. Aller, (Co-I)                                                                                     | 26 orbits, 2026 |
|                   | <b>GEMINI/FLAMINGOS2</b> - “Galactic superwinds and their role in shaping the ultra-strong metal absorbers”, (Co-I), PI: Varhsa Kulkarni                                            | 20h, 2025       |
|                   | <b>JWST/MIRI</b> – Cycle 3 - “Mining for DIAMONDS (Dust, Ice, And MOlecules iN Distant Galaxies) over the Past $\sim 7$ Billion Years” (Co-I), PI: V. Kulkarni                      | 25h, 2024       |
|                   | <b>JWST/MIRI</b> – Cycle 3 - “Probing the Dust in the Diffuse ISM in Galaxies over the Past $\sim 8$ Billion Years with Spectroscopy of Background Quasars” (Co-I), PI: V. Kulkarni | 25h, 2024       |
|                   | <b>VLT/ESPRESSO</b> – “Uncovering the Fossil Record of Chemical Evolution with Isotopic Abundance Ratios at Cosmic Noon” (Co-I), PI: V. Kulkarni                                    | 18h, 2023       |
|                   | <b>VLT/ESPRESSO</b> – “ESPRESSO Observations of a Unique CO Absorber: Paving the Way to Fundamental Physics and Cosmology with ELT-ANDES” (Co-PI), PI: S. Lopez                     | 18h, 2022       |
|                   | <b>LBT/PEPSI</b> – “Neutral Gas in and around Nearby Galaxies Mapped by MaNGA” (Co-I), PI: J. Hill                                                                                  | 10h, 2022       |
|                   | <b>VLT/X-shooter</b> – “Probing the Cold and Dense Interstellar Medium at High Redshifts Using Saturated H <sub>2</sub> Absorption Systems” (Co-I), PI: S.A. Balashev               | 16h, 2015       |
|                   | <b>VLT/UVES</b> – “Molecular Hydrogen Absorption Systems at High Redshifts: Physical Conditions in the Interstellar Medium at the Early Universe” (Co-I), PI: D.A. Varshalovich     | 9h, 2014        |

## Research Collaborations

**University of South Carolina (USA):**  
 Varsha Kulkarni (Dust and Gas in Distant Galaxies)  
 Mehrnoosh Tahani (Galactic Magnetic Fields, Dragonfly Polarimetry)

**University of Wisconsin-Madison (USA):**  
 Matthew Bershadsky (Integral Field Spectroscopy of Galaxies)

**Georgia State University (USA):**  
 Monique Aller (Dust in distant Galaxies)

**University of North Carolina Asheville (USA):**  
 David Wake (Integral Field Spectroscopy of Galaxies)

**Space Telescope Science Institute (USA):**  
 Karl Gordon (Galactic Dust Extinction)

**Presbyterian College (USA):**  
 Frances H. Cashman (High Galactic latitude gas and dust.)

**Macquarie University (Australia):**  
 Tayyaba Zafar (Extragalactic dust extinction.)

**European Southern Observatory (Germany/Chile):**  
 Pasquier Noterdaeme (Molecular absorbers)  
 Celine Peroux (Integral Field Spectroscopy of Galaxies)  
 Annalisa DeCia (Isotopic abundances, Dust depletion and extinction)

**Ioffe Institute (Russia):**  
 Sergey Balashev (Molecular absorption systems, ISM gas models)

**Inter-University Centre for Astronomy & Astrophysics (India):**  
 Raghunathan Srianand (Molecular absorption systems)

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Software         | Developer of publicly available Python packages for dust-extinction modeling, ISM physical-condition analysis, custom procedures for HST COS/STIS, JWST MIRI MRS pipelines ( <a href="https://github.com/slavaklimenko">github.com/slavaklimenko</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |
| Awards           | Best scientific work of the year, <i>Astronomy Letters Journal</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2020    |
|                  | Best scientific work of the year, <i>Ioffe Institute</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2017    |
|                  | Invited talks at conferences and institutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Several |
| Publication List | <p><b>Submitted:</b></p> <ol style="list-style-type: none"> <li>1. <b>Klimenko, V. V.</b>, Aller, M. C., &amp; Kulkarni, V. P., "Mining for Diamonds in a Lensed Galaxy <math>\sim 7</math> Billion Years Ago," <i>submitted to Nature</i>, 2026.</li> </ol> <p><b>Published:</b></p> <ol style="list-style-type: none"> <li>1. <b>Klimenko V.V.</b>, Aller M.C., &amp; Kulkarni V.P., "Probing Dust Composition in Distant Galaxies with JWST Mid-IR Spectroscopy of Quasars with Foreground 2175 Å Absorbers II. Measurements of Grain Composition and Extinction Properties," <i>ApJ</i>, 2026, 1003, 129.</li> <li>2. <b>Klimenko V.V.</b>, Aller M.C., &amp; Kulkarni V.P., "Probing Dust Composition in Distant Galaxies with JWST Mid-IR Spectroscopy of Quasars with Foreground 2175 Å Absorbers I: Methodology," <i>ApJ</i>, 2026, 998, 299.</li> <li>3. Baryshnikov I.A., Shenyavskiy V.A., <b>Klimenko V.V.</b>, Ivanchik A.V., "Modification of the method for measuring the CMB temperature based on the Sunyaev–Zeldovich effect," <i>Technical Physics</i>, 2024, 69, 1922.</li> <li>4. <b>Klimenko V.V.</b>, Balashev S.A., Noterdaeme P., Srianand A., "Excitation of CO molecules in diffuse gas over cosmic history," <i>MNRAS</i>, 2024, 533, 1367.</li> <li>5. Kosenko D.N., Balashev S.A., <b>Klimenko V.V.</b>, "Cold diffuse interstellar medium of Magellanic Clouds: II. Physical conditions from excitation of C I and H<sub>2</sub>," <i>MNRAS</i>, 2024, 528, 5065.</li> <li>6. <b>Klimenko V.V.</b>, Kulkarni V.P., Wake D.A., Poudel S., Bershadsky M.A., Peroux C., Lundgren B., "The Baryonic Content of Galaxies Mapped by MaNGA and the Gas Around Them," <i>ApJ</i>, 2023, 954, 115.</li> <li>7. Kurichin O.A., Kislitsyn P.A., <b>Klimenko V.V.</b>, Balashev S.A., Ivanchik A.V., "A new determination of the primordial helium abundance using the analyses of H II region spectra from SDSS," <i>MNRAS</i>, 2021, 502, 3045.</li> <li>8. <b>Klimenko V.V.</b>, Balashev S.A., "Physical conditions in the diffuse interstellar medium of local and high-redshift galaxies: measurements based on the excitation of H<sub>2</sub> rotational and C I fine-structure levels," <i>MNRAS</i>, 2020, 498, 1531.</li> <li>9. <b>Klimenko V.V.</b>, Ivanchik A.V., Petitjean P., Noterdaeme P., Srianand R., "Estimation of the Cosmic Microwave Background Temperature from Atomic C I and Molecular CO Lines in the Interstellar Medium of Early Galaxies," <i>Astronomy Letters</i>, 2020, 46, 715.</li> <li>10. <b>Klimenko V.V.</b>, Ivanchik A.V., "The measurements of the CMB temperature in diffuse interstellar medium of the Milky Way and high-redshift galaxies based on excitation of C I fine-structure and H<sub>2</sub> rotational levels," <i>Journal of Physics: Conference Series</i>, 2020, 1697, 012013.</li> <li>11. <b>Klimenko V.V.</b>, Petitjean P., Ivanchik A.V., "Observational estimate of the partial covering probability of quasar emission regions by distant H<sub>2</sub> absorption clouds," <i>MNRAS</i>, 2020, 493, 5743.</li> <li>12. <b>Klimenko V.V.</b>, Ivanchik A.V., "Influence of Radiative Pumping on the HD Rotational Level Populations in Diffuse Molecular Clouds of the Interstellar Medium," <i>Astronomy Letters</i>, 2020, 46, 224.</li> <li>13. Balashev S.A., <b>Klimenko V.V.</b>, Noterdaeme P., Krogager J.-K., Varshalovich D.A., Ivanchik A.V., Petitjean P., Srianand R., Ledoux C., "X-shooter observations of strong H<sub>2</sub>-bearing DLAs at high redshift," <i>MNRAS</i>, 2019, 490, 2668.</li> </ol> |         |

14. Balashev S.A., **Klimenko V.V.**, Noterdaeme P., Krogager J.-K., Ledoux C., Ivanchik A.V., Varshalovich D.A., Petitjean P., "The cold neutral phase of the interstellar medium in high-redshift galaxies," *Journal of Physics: Conference Series*, 2019, 1400, 022030.
15. Kurichin O.A., Ivanchik A.V., **Klimenko V.V.**, "Primordial  $^4\text{He}$  abundance refinement using sample of SDSS DR15 galaxies," *Journal of Physics: Conference Series*, 2019, 1400, 022051.
16. Balashev S.A., Noterdaeme P., Rahmani H., **Klimenko V.V.**, Ledoux C., Petitjean P., Srianand R., Ivanchik A.V., Varshalovich D.A., "CO-dark molecular gas at high redshift: very large  $\text{H}_2$  content and high pressure in a low-metallicity damped Lyman alpha system," *MNRAS*, 2017, 470, 2890.
17. **Klimenko V.V.**, Balashev S.A., Ivanchik A.V., Varshalovich D.A., "Estimation of physical conditions in the cold phase of the interstellar medium in the sub-DLA system at  $z = 2.06$  in the spectrum of the quasar J 2123–0050," *Astronomy Letters*, 2016, 42, 137.
18. **Klimenko V.V.**, Balashev S.A., Ivanchik A.V., Varshalovich D.A., "A new estimation of  $\text{HD}/2\text{H}_2$  at high redshift using the spectrum of the quasar J 2123–0050," *Journal of Physics: Conference Series*, 2015, 661, 012005.
19. Ofengeim D.D., Balashev S.A., Ivanchik A.V., Kaminker A.D., **Klimenko V.V.**, "Effect of a partial coverage of quasar broad-line regions by intervening cloud," *Astrophysics and Space Science*, 2015, 359, 26.
20. **Klimenko V.V.**, Balashev S.A., Ivanchik A.V., Ledoux C., Noterdaeme P., Petitjean P., Srianand R., Varshalovich D.A., "Partial covering of the emission regions of Q 0528-250 by intervening  $\text{H}_2$  cloud," *MNRAS*, 2015, 448, 280.
21. Balashev S.A., Noterdaeme P., **Klimenko V.V.**, Petitjean P., Srianand R., Ledoux C., Ivanchik A.V., Varshalovich D.A., "Neutral chlorine and molecular hydrogen at high redshift," *Astronomy & Astrophysics*, 2015, 575, L8.
22. Ivanchik A.V., Balashev S.A., Varshalovich D.A., **Klimenko V.V.**, " $\text{H}_2/\text{HD}$  molecular clouds in the early universe. An independent means of estimating the baryon density of the Universe," *Astronomy Reports*, 2015, 59, 100.
23. Balashev S.A., **Klimenko V.V.**, Ivanchik A.V., Varshalovich D.A., Petitjean P., Noterdaeme P., "Molecular hydrogen absorption systems in Sloan Digital Sky Survey," *MNRAS*, 2014, 440, 225.
24. **Klimenko V.V.**, Ivanchik A.V., Varshalovich D.A., Pavlov A.G., "Influence of gamma-ray emission on the isotopic composition of clouds in the interstellar medium," *Astronomy Letters*, 2012, 38, 364.

## Ongoing Publications

1. Putha K., **Klimenko V.V.**, Kulkarni V.P., "Constraints on the Chemical Composition of  $z > 6$  galaxies and Population III Star Properties from JWST Quasar Spectroscopy," *ApJ*, *in prep*, 2026
2. Shaban A., **Klimenko V.V.**, Kulkarni V.P., "Survey of DLA Host Galaxies at  $z \sim 2-3$  in the KCWI Archive," *A&A in prep.*, 2026
3. **Klimenko V.V.**, Kulkarni V.P., Wake D.A., Bershadsky M.A., Peroux C., Lundgren B., "The COS-MaNGA project. Analysis of the CGM around SDSS MANGA galaxies."

## References

**Varsha Kulkarni** ; Professor, University of South Carolina  
E-mail: kulkarni@sc.edu; Phone: (803) 777 6293

**Tayyaba Zafar**; Senior Lecturer, Macquarie University  
E-mail: tayyaba.zafar@mq.edu.au; Phone: 02 9372 4809

**Matthew A. Bershadsky**; Professor, University of Wisconsin-Madison  
E-mail: mab@astro.wisc.edu; Phone: (608) 265-3392

**Sergei Balashev** ; Senior Researcher, Ioffe Institute  
E-mail: s.balashev@gmail.com; Phone: 812 292 7180

## Talks and Conferences

- 11/2025 – Tucson, AZ: Poster at Dusty Universe 2025: “Extinction Curves from UV to Mid-IR from Dust In and Around Distant Galaxies”
- 06/2025 – Princeton University, NJ, USA: “Probing Dust Composition in Distant Galaxies with JWST Mid-IR Spectroscopy of Quasars with Foreground 2175Å Absorbers”
- 04/2025 – Francis Marion University, SC, USA: “The Baryonic Content of Galaxies Mapped by MaNGA and the Gas Around Them”
- 11/2023 – Santiago, Chile: “Cool-gas metallicity distribution around nearby galaxies mapped with integral field spectroscopy”
- 12/2019 – IKI RAS, Moscow, Russia: Poster: “Radiative pumping of HD rotational levels in the diffuse ISM”
- 12/2019 – SAO RAS, Russia: Poster: “Interstellar UV radiation field in high redshift galaxies probed by Damped Lyman Alpha systems”
- 04/2019 – Pushino, Russia: “Physical conditions in the diffuse ISM probed by excitation of H<sub>2</sub> rotational levels”
- 12/2018 – IKI RAS, Moscow, Russia: “Physical conditions in translucent clouds at high redshift probed by excitation of CO rotational levels”
- 12/2017 – IKI RAS, Moscow, Russia: “Measurements of the CMB temperature in the early Universe”
- 04/2015 – Pushino, Russia: “Physical conditions of the cold phase of the neutral ISM at high redshift”
- 12/2014 – IKI RAS, Moscow, Russia: “Physical conditions in dense molecular clouds probed by the analysis of excitation of H<sub>2</sub>/C I levels in quasar spectra”
- 12/2014 – Swinburne University, Kuching, Malaysia: Poster: “HD/H<sub>2</sub> molecular clouds at high redshifts”
- 06/2014 – St. Petersburg, Russia: “The effect of partial covering of QSO emission regions by distant absorption clouds”
- 05/2013 – St. Petersburg, Russia: “Measurements of the variation of electron-to-proton mass ratio from the analysis of H<sub>2</sub> absorption systems in quasar spectra”
- 05/2011 – Catania, Italy: “The contribution of longitudinal plasmons to the thermodynamics of dense electron plasma”
- 10/2010 – St. Petersburg, Russia: “Influence of the gamma-ray emission on the isotopic composition of clouds in the interstellar medium”